Triggering Behavior Changes with Information and Incentives – An Active Traffic and Demand Management-Oriented Review

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Abstract: Traditionally, methods to reduce traffic congestion has been focused on the capacity side. An increasingly popular approach is to look at issues from the demand side to trigger travel behavior changes, so as to achieve a better balance between the need to travel a particular route at a particular time and the capacity of available facilities which is defined by FHWA as active traffic and demand management strategy (ATDM). The provisions of dynamic traffic information and incentives are two most commonly seen methods to effectively influence people's travel behavior. This chapter aims to provide a comprehensive and in-depth review of the research work that focus on triggering travel behavior changes with the provisions of dynamic traffic information and incentives via Information and Communication Technologies (ICT). Real-world system developments and current practices in adjusting travel behaviors will be summarized first, followed by the review of various behavior change aspects, including departure time choice behavior, route choice behavior, mode choice behavior and destination choice behavior. Driving behavior, which is a densely populated topic is also reviewed and summarized. The reviewed studies are summarized and compared at the end of each corresponding section.

Keywords: Travel behavior change, Information and Communication Technologies (ICT), Active Traffic and Demand Management (ATDM), incentives, behavior choices

1. INTRODUCTION

Traffic congestion happens due to demand-supply imbalance in the transportation network. Traffic flow slows down when the number of vehicles travel on the road increases or the roadway capacity decreases due to various reasons. It brings a series of issues including extra travel times for the drivers and passengers, increased fuel consumptions and greenhouse gas emissions, higher vehicular crash rates and so on. Traditionally, the focus has been on the capacity side such as building more roadway infrastructure to accommodate traffic demand with higher capacity, or applying Transportation System Management (TSM) strategies to increase capacity without adding physical infrastructure. Such strategy worked initially, but later on became increasingly expensive and unsustainable.

An increasingly popular approach is to look at issues from the demand side. For example, a strategy named Active Traffic and Demand Management (ATDM) aims at a better balance between the need to travel a particular route at a particular time and the capacity of available facilities to handle this demand efficiently. As defined by Federal Highway Administration (FHWA), ATDM is being considered as the market-ready technologies and innovative operational approaches that are becoming available for managing traffic congestion within the existing infrastructure (Luten et al. 2004). It refers to the wide-range and diversified characteristics of traffic management strategies deployed by transportation agencies that focus on using advanced technologies, available information, and valid methodologies, in order to monitor and manage the dynamic traffic conditions actively, and to influence people's need and intention to travel as well as their associated travel behavior, for the purpose of promoting efficient use of existing roadway systems and better handling of the vehicle demand (Hu 2013). ATDM offers significant potential for reducing traffic congestion without the need to build additional lanes or infrastructure. The vision for ATDM research is to allow transportation agencies to increase traffic flow, improve travel time reliability, and optimize available capacity throughout the transportation network.

The emphasis of ATDM on travel behavior changes suggests a paradigm shift from the previous-popular term Advanced Traveler Information Systems (ATIS). Real-Time Traveler Information Systems or ATIS

(such as Google, Waze, 511, etc.) provide pre-trip and/or enroute information allowing travelers to quickly assess and react to unfolding traffic conditions. The basic design concept is to present generic current information to travelers, leaving travelers to react to the information their own way. This "passive" way of managing traffic by providing generic traffic information makes it difficult to predict outcome and may even incur adverse effect, such as overreaction, which may lead to herding effects. Here, herding effects, or sometimes refers to "herd behavior", describes how individuals in a group can act collectively without centralized direction (Hu et al. 2017). To be specific, if a particular road segment is identified to be free flowing, ATIS will recommend this route to every driver around. If they all follow the navigation device, after a while, this road segment will become congested, and drivers may find themselves stuck in traffic and sitting in the long queue again. On the contrary, ATDM aims to effectively influence people's travel demand by providing more travel options including departure time options, route choices and travel modes options, coordinating between travelers, improving people's driving behavior and smoothing traffic flow, and reducing the need for travel. Such behavioral adjustment approach has been receiving continual attention from both academic research and real-world practice, and this chapter aims to summarize the most important work to date.

There's also significant difference between ATDM and the traditionally defined Travel Demand Management (TDM). Most current TDM programs such as vanpooling, ridesharing, or transit focus on managing travel demand of specific groups of commuters but are limited in effectively managing demand for automobile drivers, who are unable or unwilling to participate in such programs. Another aspect of TDM worth noting are travel pricing strategies, such as congestion pricing, HOV to HOT conversion, and parking pricing, which aim to influence travelers' behavioral changes by imposing monetary penalties on commuters. Although some of these strategies have proven to be, to a certain extent, effective in changing travelers' behavior while increasing revenue generation, this "stick" approach remains controversial on issues like socio-economic equity and capacity utilization efficiency. Additionally, in practice, imposing road pricing is controversial and insight is lacking in key domains which could lead to different outcomes than those predicted by economic theory (Ben-Elia et al. 2011). ATDM goes beyond traditional TDM by

not only not limiting itself to those non-auto travelers, but also including non-pricing strategies to effectively manage demand and subsequently the transportation system. As such, ATDM can be viewed as a conceptual merge between TDM, ATIS and TSM.

The provisions of dynamic traffic information and incentives are two most commonly seen methods to effectively influence people's travel behaviors in an ATDM system. With the use of Information and Communication Technologies (ICT) emerges among the population, more and more traffic information from various channels are collected and shared with public and social networks instantaneously. The real-time travel time and accidental information can help travelers plan ahead and adjust their departure times based on personal preferences and needs. For example, a commuter can leave later to avoid sitting in the queue caused by an accident, or a shopping trip can be canceled or rescheduled because of a foreseen delay. On the other hand, incentives, which could take forms of monetized lottery/rewards, free transit ticket, travel feedback (such as gas saved), or environmental contributions, is increasingly applied together with the state-of-art ICT in making a fundamental paradigm shift from the traditional passive travel information provision to an active traffic demand management tool, by influencing people's need and intention to travel as well as their associated travel pattern. With the impact of incentives, the information has more meaningful suggestions for the traveler and helps the decision making process about what time to leave, what mode to use, which route to take, or even to travel or not (Hu et al. 2019).

This chapter aims to provide a comprehensive and in-depth review of the research work that focus on triggering travel behavior changes with the provisions of dynamic traffic information and incentives via ICT. Real-world system developments and current practices in adjusting travel behaviors will be summarized first in section 2, followed by the review of various behavior change aspects, including departure time choice behavior (section 3), route choice behavior (section 4), mode choice behavior and destination choice behavior (section 5). Driving behavior, which is a densely populated topic is also reviewed and summarized in section 6. Section 7 concludes this chapter.

2. SYSTEMS DEVELOPMENT AND CURRENT PRACTICES

Over the past few decades, before ICT was in place to support traveller's decision makings or collect data to enable research, some behavioral experiments were implemented to influence people's travel behavior decisions. For example, the University of Washington at Seattle engaged in the U-Pass program, which started in the 1990s and aimed to alleviate the traffic problem both around the campus and within the city. The program provided an array of transportation options, including transit service, preferential parking, consumer discounts, and rideshare matching. The benefits of this program had been significant: the building of 3,600 new parking spaces had been avoided, 86% user satisfaction had been reported, and a reduction of 33% had been seen in parking permit purchases, plus 18% less in traffic counts during morning peak hour in 2002, compared with the data in1983 (University of Washington 2002). Lee County, FL used variable Bridge Tolls to manage the traffic congestion, and was able to spread traffic away from the peak period. Over the peak hours, the tolls on two principal bridges were raised from 75 cents to one dollar, and for the off-peak time a 50-cent discount was provided. With the $9.7 million grant from the Federal Highway Administration and another $7 million "emergency revenue reserve," 5% of traffic was shifted from peak to off-peak time periods, although as many as half of respondents indicated they always or sometimes considered the discounts prior to making a trip across one of the bridges (Burris 2001). Similar dynamic pricing strategy can also be found in Seattle, Los Angeles, San Diego, Atlanta, where dynamic pricing of High Occupancy/Toll (HOT) lanes and incentives for High Occupancy Vehicle (HOV) usage were provided.

The Neighborhood Eco (NECO) Pass program started in Denver, Colorado in 1997 to reduce single-occupancy vehicles (SOV) and had grown to include 45 neighborhoods and more than 6,500 households. The NECO Pass was a yearly transit pass option that was primarily used by neighborhoods in the City of Boulder. The lower price and convenience of a NECO Pass was intended to incentivize residents to shift their mode of travel to transit. According to the report (National Research Center 2016), SOV trips had declined by 7.7 % since 1990. The transit-oriented development (TOD) in Hillsboro, in the Portland

Metropolitan area in Oregon, had made it one of the most liveable communities. It featured a pedestrian friendly environment and mixed-use development in the station area which made it possible for the residents to walk to the light rail station. New tenants were offered a one-year transit pass free of charge. Due to this program 83% of residents reported they used transit in May 1999, and transit usage for commuting had increased 22%. It was also reported that 85% of respondents stated they had reduced needs to drive because of the program (Podobnik 2002). Zion National Park program started in 2000, and the propane-powered, shuttle system vehicles became the only vehicles permitted in the park during the peak season. Visitors were encouraged to park outside and used the shuttle free of charge to access the national park. Shuttle boarding reported for year 2002 was 2.35 million, and about 75% of the visitors rode the Zion Canyon shuttle. Besides the benefits from traffic and parking aspects, the local government benefited from this program as well, not only from the reduction in infrastructure cost but also from the economic growth (US NATIONAL PARK SERVICE 2003). In 2000, One Less Car program was launched in Seattle WA and continued through 2002. The program offered households information financial incentives to help them reduce their car use for all their trips. In total, 86 households participated in the study for a period of 6 to 9 weeks, ranging from households with children, to singles, to empty-nesters. Participating households received a weekly stipend of $85 to compensate them for the extensive data they recorded and the republic resource they saved. At the end of the study, participants reduced their drive-alone car miles by 27%, and the use of alternate modes of transportation increased more than 20% (Lee et al. 2013). In 2003, the NuRide (renamed as Agile Mile) program was established in the US and offered through State and local governments and metropolitan planning organizations. NuRide program is a reward program for commuters who were encouraged to take non-motorized modes and car/vanpooling. Users who signed up received points for each green trip they record via an online portal. Points can be redeemed for coupons for restaurants and retail shops, services, movie tickets, etc. Since established, NuRide had built a presence in nine States. For example, in the Commonwealth of Massachusetts, NuRide indicated that participants in the State have saved more

than 175 million miles of driving, more than 8 million gallons of gasoline, nearly $97 million in travel-related costs, and nearly 79,000 tons of emissions since its launch in 2010 (Bauer et al. 2018). In 2001 in Japan , a survey of 335 drivers was carried out before and during an 8-day temporary freeway closure. The results illustrated that the frequency of switching to public transport during the closure was inversely related to the frequency of automobile commuting before the closure. Furthermore, habitual drivers tended to have negative beliefs about using public transport, and that the negative beliefs were corrected by actual experiences of using public transport (Fujii et al. 2001). In 2003, an experiment with respect to 43 drivers was carried out during an eight-day temporary freeway closure in Osaka, Japan, where 23 travelers were offered a one-month free bus ticket, but the free ticket was not given to the other 20 drivers in a control group. The result showed that drivers who received a one-month free bus ticket in the experiment used the bus more frequently after the intervention. The increase was 20% higher than the frequency of bus use before, although it turned out in this experiment that these behavior changes were not permanent (Fujii and Kitamura 2003).

In Germany, 2003, a longitudinal study investigated the effects of an intervention on increased bus use among college students by offering the incentives of a pre-paid bus ticket. A two-wave panel design with a 1-year interval between waves was deployed. The intervention was found to influence attitudes toward bus use, subjective norms, and perceptions of behavioral control. The bus usage had increased by 35% and trips finished by car decreased by 13%, which indicated the incentive given to the travelers can influence their travel behavior as compared to their previous habitual behavior (Bamberg et al. 2002). In a different experiment, effects of an intervention (combination of information and a free public transport ticket) in changed decision context (moving to a new residence) on travel mode choice by car users was examined. The participants consisted of 241 people planning to move within the next 6 months to Stuttgart, Germany. A lottery ticket with attractive money prizes was used as an incentive to participate. The results showed that the intervention influenced attitude, subjective norm, and perceived behavioral control, and consistent with Ajzen's theory of planned behavior (Ajzen 1991), these were the main causes of the change of travel mode (Bamberg et al. 2003).

Many other behavioral experiments have been implemented worldwide to adjust traveler's behavior decisions. For example, in Melbourne, Australia, an early bird ticket program was proposed to alleviate the rail overcrowding issue during peak hours. Free rail fares were provided for rail travelers completing their trips before 7:00 a.m. as incentives to shift demand from the peak to pre-peak trains in order to relieve the overcrowding problem. The cost of free rail program was nearly $6M mainly in lost fares and around 8,000-9,000 passengers used the ticket on weekday. It ended up that 23% of ticket users (2,000-2,600 each week) had shifted from the peak hour by an average of 42 minutes; the ticket use was increased by 1.7% while overall rail usage was stable. After introduction of the early bird program, pre 7:00a.m loads increased 41%, reducing pressures for purchase/operations of new peak trains saving 2.5–5.0 trains (2008) to a forecast saving of 8.05 trains in 2038. Financial analysis suggested the savings would substantially cover the financial costs of the free fares and that benefits would increase over time (Currie 2011). Two fare policies, off-peak discounts and volume rebates, had been applied in South East Queensland attempting to alter passengers' travel behavior. The results showed that passengers' boarding times were not significantly altered under current off-peak discounts. However, volume rebates provided incentives to encourage more use of public transport (Yen et al. 2015).

In India, the INSTANT (for the Infosys-Stanford Traffic project) was conducted in Bangalore for six months from Oct 6, 2008, to April 10, 2009. There were about 14,000 commuters participating in the experiment, and bus data was collected from 9,000 of them traveled in one of over 200 buses chartered by Infosys. The scheme awarded credits each day to employees based on their arrival times and each week the cumulative number of credits of each commuter was used by an algorithm to choose commuters who will win monetary rewards. The algorithm had three components: credit allocation, weekly reward draws and credit deduction. Every working day a commuter were awarded credits according to his/her swipe-in time: 1.5 credits for a swipe-in before 8 a.m., 1 credit for a swipe-in between 8 a.m. and 8:30 a.m. and no credit after 8:30 a.m. At the end of each week, the algorithm determined the level at which a commuter has qualified. After each reward draw, credits were deducted from both winners and non-winners. The results showed that the number of commuters arriving in each time segment had almost doubled over the

period of the experiment and the number of buses that were over capacity had significantly reduced. The project attracted a large number of commuters to travel at off-peak hours, leading to a advancement of pickup times in the bus schedule (Merugu et al. 2009).

Such behaviour experiments can also be found in Singapore. In June 2013, a "Travel Early, Travel Free," campaign was launched by Singapore's Land Transport Authority to alleviate subway crowding, which offered free trips to those who exited from one of the 16 (extended to 18 following the completion of a new subway line in 2014) city-center subway stations before 7:45 a.m. on weekdays. Commuters who exited within the next 15 minutes received a 50-cent discount. Preliminary data indicated a reduction of 7%–8% in the number of commuters during the morning peak window (Land Transport and Authority 2014). Similarly, the randomized controlled trials on Singapore's subways were implemented in August, 2014 and lasted for 10 weeks. Commuters who registered received a full rebate of their fares if they exited fare gates before the start of morning peak-travel time. A difference-in-differences analysis revealed that the effect of the free-trip offer was modest but robust during the promotional period: trips were 5.5% on average more likely to end before the morning peak time (Yang and Lim 2018). The objective of both campaigns was to alleviate peak-time congestion commuters to travel pre-peak.

With the rapid advancements of ICT, systems have been developed to influence traveler's mode choice or departure time choice behaviors. Such systems could include smartphone apps, software, or frameworks that allows system operators to provide means such as dynamic traffic information or monetary incentives to travelers, with the goal of influencing their trip-making decisions. For example, as early as two decades ago, Georgia Power developed a software tool named "SmartRide" to track carpool, vanpool and transit use information and direct the information to its employees, with the goal of getting as many cars off the roads as possible during the 1996 Olympic Games in Atlanta. The company also provided fleet vehicles to its employees to run errands or go to meetings during the day, so those who used alternative transportation to commute to and from work would not feel trapped in the office during the day. This program successfully reduced 1.2 million Vehicle Miles Traveled (VMT) each month, with 13% of employees carpooling or vanpooling, 15% working compressed or flex-time schedules, 5% using transit

and another 5% doing telework (FHWA 2006). Over the last decade, carpool apps, including ad-hoc causal carpooling, commercial ridesharing and other forms became very popular. Examples include Carma Share, Zimride, Waze Carpool, Rideamigos, BlaBlaCar, Uber, Lyft, iCarpool, GoKid and many others with varying penetration rates which is one of the key elements determining their effectiveness. However, due to page limitation they cannot be reviewed in-depth individually.

In 2006, the Spitsmijden project was launched in Netherlands, which involved rewarding of commuters for avoiding travelling by car during the peak hours using automated vehicle identification technology. The test area was the Dutch A12 motorway corridor from Zoetermeer towards the Hague. Upon registration, one of two types of reward were chosen by the recruited 500 participants. The first type of reward was an amount of money ranging from 3 to 7 EURO for each morning rush hour that the participant avoided. The second type comprised savings towards keeping a "Yeti" smartphone at the end of the trial (AENEAS 2010). The results reported of the study indicated that positive incentives were able to reduce the amount of peak traffic of the participants by about 60% and travelers mainly responded to the incentives by shifting their car trips to the periods before and after the peak period (Ettema et al. 2010). It was also reported that travelers returned to the peak period when the incentives ended. Knockaert et al. (2012) also confirmed that a reward can be used as an effective policy instrument, and that the participant's behavior implied that the shadow prices of schedule delay were close to constant over time which was in line with the classic assumptions in the literature. Shifting departure time was found to be a more important behavioral response to policies for congestion relief than a modal shift or teleworking. Ben-Elia and Ettema (2011) further found out that although the reward influence the magnitude of change, choosing how to avoid the rush-hour, which includes driving at other times, switching to another mode of transport or working from home, was not determined by the reward, but rather by different factors relating to the participants and their particular situations. More research work based on Spitsmijden will be discussed later in the corresponding behavior analysis sections

The information- and reward-based travel demand management strategies started to be implemented on the smartphones. In 2013, Metropia (www.metropia.com), for example, used traffic prediction and

coordinating technologies combined with user rewards to incentivize drivers so as to balance the traffic load on the network, and reduce traffic congestion. Screenshots from the app are presented in Figure 1**Error! Reference source not found.** to describe how the app works. Drivers use the app to plan a trip, and the system provides travel time prediction for every 15 minutes and varying levels of incentives which are based on the traffic condition with the goal of attracting users to the less congested departure times. The reward points are determined based on the departure time and congestion levels of the road. If the departure time and route selected by user is heavily congested, the points will be lower, and if the users are willing to leave during the non-peak hours, they can get more points. Once the user decides on the time and route to travel, the system will verify if the user indeed leaves at the time promised and follows the desired route, and if yes, the trip is validated and the points will be added to the user's account. The user can keep using the app to accumulate the points. The points carry monetary value, and can be used to redeem freebies such as Amazon and Starbucks gift cards. (Hu et al. 2019). The goal of the app and its backend algorithms is to influence its users to reduce demand during peak hours for congested routes (Arian et al. 2018). Under the hood is a set of transportation system models and machine learning algorithms to estimate the dynamic traffic conditions in real-time and further perform traffic predictions (An et al. 2016, Hu et al. 2017), with the goal of improving real-world traffic conditions from user-equilibrium (UE) or non-UE towards system optimum (SO) (Hu and Chiu 2015, Hu et al. 2017). A small-scale pilot field study was carried out in Los Angeles, CA together with Los Angeles Department of Transportation (LADOT) since April 2013 for a total of 10 weeks, the results showed that travel behavior changes were observed and 20% travel time could be saved for people who changed their travel behavior (Hu et al. 2015). The system is currently available in a number of cities in the US, and is expected to expand to a few new cities in the near future.

[Insert Figure 1 Metropia smartphone app workflow here]

In 2017, The Bella Mossa program was launched in Bologna, Italy and used a system of gamification and rewards to encourage people to choose environmentally friendly transportation modes such as walking, bicycling, or transit. They used a GPS-based smartphone application to track users' sustainable trips and

then provide various rewards and incentives for their choices. During the initial 6-month launch of Bella Mossa program, 15,000 people actively participated in the incentive program, which resulted in an estimated 3.7 million kilometers (2.3 million miles) of sustainable travel and 900,000 sustainable trips (Bauer et al. 2018).

Another smartphone-based application "incenTrip" was developed by researchers from the University of Maryland to study how to show the right travelers the right incentives delivered at the right time, with the goal of maximizing energy and travel time savings. The app aims to make commuting easier, faster and cheaper by encouraging commuters to make smart choices and reward them when they do. incenTrip app uses real-time data to predict travel times on multiple modes of transportation, offering commuters options for getting where they want to go and offering users points that can be redeemed for prizes. The app had 35,000 registered users (accounts for approximately 0.4% of the total 9 million population in the region) as of April, 2018 per the Washington Post and is designed for use in the Washington and Baltimore regions (The Washington Post 2018). The project was initially funded by the Traveler Response Architecture using Novel Signaling for Network Efficiency in Transportation (TRANSNET) program by U.S. Department of Energy's Advanced Research Projects Agency (ARPA-E) in 2018 (ARPA-E 2018). The key goals of the app are through the development of integrated, personalized and real-time traveler information and incentive technologies, employ behaviorally sound and computationally efficient agent-based modelling (ABM) system to simulate the entire transportation system in real-time, and ultimately incentivize more energy-efficient travel and mobility decisions. Figure 2 illustrates the system architecture and workflow of the app. Through a demonstrative case study for a large-scale transportation system of the D.C. – Baltimore region, the capability of the proposed technology is highlighted with significant system-level energy savings, reasonable insights on individual travel behavior responses, as well as superior computational efficiency (Xiong et al. 2018). Under the hood is a new model that integrates an activity-based travel demand model (ABM) with a dynamic traffic assignment (DTA) model. Specifically, the integrated model includes InSITE, an ABM developed for the Baltimore Metropolitan Council, and DTALite, a mesoscopic DTA model. The integrated model

simulates the complete daily activity choices of individuals residing in the model region, including long-term choices, such as workplace location; daily activity patterns, including joint household activities and school escorting; activity location choices; time-of-day choices; mode choices; and route choices (Zhang et al. 2018).

[Insert Figure 2 (a) incenTrip architecture, and (b) incenTrip workflow here]

Parking is another populous area for researchers and practitioners to offer real-time information and provide cost comparison with the goal of influencing driver's parking decisions. In April 2012, an incentive program named Congestion and Parking Relief Incentives (CAPRI) was launched in Stanford University in Palo Alto, California. Credits for avoiding peak parking hours were awarded to the travelers who enter and exit the main Stanford University campus by car at designated off-peak hours Monday through Friday. The goal of the program was to manage the parking problem through dynamic parking pricing strategies to reduce peak-hour traffic. The credits can be used to win random cash rewards from $2 to $50 and the rewards can be accumulated and disbursed monthly via Stanford's payroll or through bank deposits (Stanford University 2012). 3082 commuters registered for the automobile program and 975 registered for the working and biking incentives using the smartphone. A total of 10,200 permit-holding car commuters at Stanford were invited. The results showed that the rewards in CAPRI were effective in encouraging commuters to drive during off-peak hours, and to walk and bike (Yue 2014, Yue et al. 2015). Aside from this kind of academic researches, commercial apps also emerged quickly in the market, such as Parking Panda, Parkmobile, Parker, ParkWhiz, and many others. The design principal of these type of apps is to offer drivers with the available parking information (and some with pricing information) on the smartphone, so as to assist drivers with their travel and reduce the time wasted in cruising to find a parking space, and the associated fuel consumption and negative environmental impacts. A comparison of the reviewed system development and current practices is summarized in Table 1 below, in chronological order. The comparison is based on whether a real system/product had been developed and released (system development), if dynamic traffic information is provided (information), are

incentives or pricing strategies (carrot versus sticks) used, the scale of experiment such as number of users and area of implementation, and what behavior choice(s) does it target at.

[Insert Table 1 Comparison of systems development and behavior experiments here]

3. DEPARTURE TIME CHOICE BEHAVIOR

As one of the fundamental travel behavior dimensions, departure time choice is a densely populated area that has received lots of research attentions in the past. As reviewed in section 2, majority of system development and behavior experiments efforts also focused on triggering department time change. Due to the high volume in literature, this section tries to review the most relevant and significant ones, specifically, those focused on answering the question of what it takes to trigger departure time changes. These literatures will be summarized according to different type of trips investigated, i.e. home-based trips, work and shopping trips, peak-hour trips and 24-hour trips. Research on departure time evolution and learning is also summarized in this section. The reviewed studies are summarized and compared at the end of this section. It is generally agreed that a number of factors such as work schedule flexibility, mode, occupation, income, age, transportation level of service and others all influence departure time choice.

3.1. Home-based trips

The effect of Advanced Traveler Information Systems (AITS) on travelers' route and departure time changes of home-based trips were evaluated in (Khattak et al. 1995). Survey data was collected from downtown Chicago automobile commuters, as when this early study was conducted, the data collection was challenging without real ICT in place. The survey mainly investigated the home-to-work trips and more than 60% of the respondents had used traffic information to modify their travel decisions. Analysis showed that respondents were more likely to change their departure times if they perceived traffic reports to be accurate and relevant, and frequently listened to traffic reports. The implication for Advanced Traveler Information Systems were that they may be designed to support both enroute and pre-trip decisions. ATIS performance, measured in terms of accuracy, relevance and timeliness would be critical

in the success of such systems. Further, near-term prediction of traffic conditions on congested and unreliable routes (where conditions change rapidly) and incident durations was desirable.

The joint decisions of trip timing and mode choice for Home-Work-Home trips and the correlation between discrete and continuous decisions were investigated in the Greater Toronto Area. The joint model, with a multinomial logit model for mode choice and a continuous time hazard model for trip timing, allowed for unrestricted correlation between the unobserved factors influencing these two decisions. Strong correlations between unobserved factors influencing mode choice and trip timing were found across all occupation groups, and it proved that it sufficiently captured the peak spreading phenomenon (Habib et al. 2009, Habib 2013).

The effect of constraints in work and in other daily trips/activities on the willingness to shift departure time and the willingness to pay for reducing travel time and travel delay was investigated in (Thorhauge et al. 2016). Detailed data on the full 24-hour out-of-home activities and on the constraints for each of these activities was collected in a survey. A stated preference experiment to infer preferences on departure time choice was built, and a mixed logit model was estimated, based on the scheduling model, to account for the effects of daily activity schedules and their constraints. The results showed that measuring flexibility in terms of work start time or constraints at work did not provide exactly the same information, and if the presence of constraints were neglected, it would generally lead to biased value-of-time estimates. The shift in the departure time, especially towards a late departure time, was strongly overestimated (the predicted shift was more than double) when the effect of non-work activities and their constraints was not accounted for.

A group of researchers from the University of Maryland inferred travel decision with a mixed Bayesian network (BN) in (Zhu et al. 2017). A real-world case study was conducted by using the regional travel survey data (including 24-hour out-of-home trips) for a two-dimensional decision modeling of both departure time choice and travel mode choice. The results demonstrated that mode and departure time were not independent of each other, and the variance of departure time choice across different social

demographical variables was not as significant as the variance of mode choice. The proposed two-dimensional mixed BN provided better accuracy than decision tree models and nested logit models. Heterogeneous travelers' behavior toward trip mode and departure time choices was investigated by considering those choices as a joint decision in (Shabanpour et al. 2017). The survey data contained 210,000 home-based trip observations and detailed information for the travelers. First, a two-step clustering algorithm was applied to classify travelers into six distinct clusters to account for the heterogeneity in their decision-making behavior. Then, a joint discrete-continuous model was proposed for each cluster, in which the travel mode and departure time were estimated by a multinomial logit and a log-linear regression model, respectively. The results highlighted that travel time and household size significantly affect departure time decision. The increase in travel time resulted in earlier departure times and more vehicles in the household leaded to later departure time.

3.2. Work and shopping trips

Using data collected as part of a larger survey on the consequences of congestion charging on travel choices in the city of Edinburgh, multinomial logit models were calibrated to investigate the influence of non-work activities and socio-economic characteristics on scheduling flexible for work trips. The results showed that non-work activities and work schedule flexibility had an impact on departure time choice for the journey to work, which meant that even for those with flexible work schedules, but with other non-work commitments, the timing of their work trip may not be so flexible (Saleh and Farrell 2005). In another research, in order to test the effect of flextime in reducing morning peak-hour traffic, a multinomial logit model was built which divided morning departure times as pre-peak (0:00 a.m. to 6:30 a.m.), peak (6:30 a.m. to 8:30 a.m.) and post peak (8:30 a.m. to 12:00 a.m.). The results demonstrated a correlation between having flextime and traveling in post-peak hours, and also proved that commuters with different characteristics may react diversely to flextime (Yang et al. 2016).

Bhat and Steed proposed a continuous-time hazard duration model for urban shopping trip departure time choice. Shopping trip data from the activity survey collected in the Dallas-Fort Worth metropolitan area

was used. The effects of individual and household socio-demographics, employment attributes, and trip-related characteristics on departure time choice were presented and discussed. Application of the continuous-time model to forecast temporal shifts in shopping trip-making due to changes in socio-demographic characteristics and trip-chaining behavior was demonstrated (Bhat and Steed 2002). Yang et, al. also studied the joint choice of residential location, travel mode, and departure time based on random utility maximization theory with Cross-Nested Logit model and traditional NL models. A number of exogenous variables including house price, travel time, travel cost, and factors depicting the individual socio-economic characteristics were defined, and the model choice sets were the combination of residential location subset, departure time subset, and travel mode choice subset. The model parameters were estimated using Beijing traffic survey data, and the direct and cross elasticity were calculated to analyze the change of alternatives probability brought by factors variation. Estimation results showed the Cross-Nested Logit model outperformed the three kinds of NL model, and decision makers would change first their departure times, then their travel modes, and finally their residential locations, when exogenous variables alter (Yang et al. 2013).

3.3. Peak-hour and 24-hour trips

Departure time choice models were proposed based on the notion of a latent preferred arrival time using the peak-avoidance data of the Dutch "Spitsmijden" experiment, which involved the use of rewards (money or Smartphone credits) for encouraging drivers to avoid commuting during the morning peak-hours. Three models were estimated to identify car drivers' choice of departure time when rewards were provided: a logit model, a mixture model and a latent class model regarding arrival time preference. The results indicated that the rewards, both monetary and in-kind (a Yeti smartphone) had a substantial effect on decreasing peak travel. Furthermore, in the latent-class specification, women were more likely to have a preferred arrival time compared to men (Ben-Elia et al. 2010).

Psychological factors were derived to build departure time hybrid choice model based on the theory of Planned Behavior, estimated them based on structural equation modelling, and included them into a

discrete choice model. The psychological factors were measured based on an online questionnaire addressed to car commuters to the city center of Copenhagen in the peak-hour. The effects of the psychological factors were different depending on framework conditions: for people with fixed starting times at work, the intention to arrive at work on time (as estimated by subjective norm, attitude, perceived behavioral control) had the strongest effect; for people with flexible working hours, the attitude towards short travel time was most relevant (Thorhauge et al. 2016).

The direct effect of the intention on the actual behavior was explored in (Thorhauge et al. 2017), which used data from Danish commuters in the morning rush hours in the Greater Copenhagen area. The role of intention as mediator between several latent effects and the behavior was studied. Attitude, social norms, and perceived behavioral control were found to affect the intention to behave in a given way. A hybrid choice model was applied to study the departure time choice. The results showed a significant effect of the intention to arrive at work on time on the departing time choice, and also a significant effect of the lower level mediators (such as attitude, intention, attitude, etc.) on intention. Furthermore, the attitude toward short travel time was also significant in explaining the departure time choice. Individuals who had a strong intention to be at work on time were forecasted to be less likely to reschedule their departure time.

How travelers changed their daily travel decisions based on accumulated experience was explored in (Wang 1996), in which a two-stage simulation model was established to estimate people's timing choices and timing utility over all major activities. The hazard-rate duration was used to estimate the revealed preferences of Canadians on their major daily activities. With the results of this simulation, the tradeoff between travel time and the scheduling choices was examined and a commuter equilibrium was established on a basis of activity timing utility, work start-time, and travel time. In a different research, a micro-simulation approach was applied to explore how travelers base their daily travel decisions on the accumulated experience gathered from repetitively travelling through the transport network. Individuals based their consecutive departure time decisions on a mental model, which was the outcome of a continuous process of perception updating according to principles of reinforcement learning. Individuals'

daily travel decisions were linked to the traffic simulator SIAS-PARAMICS to create a simulation system in which both individual decision-making and system performance (and interactions between these two levels) were adequately represented. The simulation results suggested that the outcomes of the learning and adaptation process may not be straightforward, but was a powerful tool in modelling the dynamics in responses over time (Ettema et al. 2005).

3.4. Departure Time Learning and Evolution

With the provision of dynamic traffic information and incentives, the day-to-day departure time choice evolution of the travelers was explored in (Hu et al. 2019), using real observation data collected from a smartphone app "Metropia". Influenced by the current information and incentives provided in the app and the comparison with the experience gained from the last trip, a transformation process of traveler's day-to-day experience on departure time from an existing habit to a new one was analyzed. A binary choice model was proposed for the shift of departure time for each repeated morning commute trip comparing with the last one, which proved that users' experience in app engagement, previous travel time saving, habitual travel time, incentives, and commute flexibility were able to trigger day-to-day behavioral change for their morning home-to-work commutes.

In a relevant research, the impact of incentives on departure time for seven main trip purposes was analyzed to further provide a personalized incentive scheme to achieve an optimal manner at the system wide level. Two consecutive incentive schemes were compared, each lasting for over a 1-year period, to understand how effectively these incentive scenarios influence the departure time of seven trip purposes. Four accelerated failure time models were conducted to explain the time to observe a departure event and the acceleration factors for various trip purposes under two incentive scenarios. The results suggested that peak hour travelers might be open to behavioral incentives that promotes non-peak hour departure time and incentives might need to be tailored based on time of the day and the purpose of the trip. Traveling for recreation, errands, and work was the most sensitive trip purposes to the incentives in the morning

peak hour. However, in the evening peak hour, traveling for other trips, errands, and school became the most sensitive trip purposes (Arian et al. 2018).

The learning process of travelers' departure time choice behavior was modeled based on the Bayesian learning theory, using the information received from social media and location-aware mobile devices as inputs. A general framework was introduced for departure time choice with information sharing via social networks. The parameters in the learning model were estimated based on an experimental data set. The agent-based approach, which was proposed to simulate travelers' choices, was applied to validate the model and examine the effect of different social network structures, in terms of both travel choices and transportation system performance (Xiao and Lo 2016). Liu et, al. validated that travelers can learn from their past travel experiences as well as the traffic forecasts offered by the transport information provider/agency in adjusting their departure time and mode choices, while the information agency learns from historical data in updating traffic forecast from day to day at the same time. When exploring the convergence of the proposed model to the dynamic bi-modal commuting equilibrium, the authors found that appropriate traffic forecast can help the system converge to the user equilibrium and user inertia might slow down the convergence speed of the day-to-day evolution model (Liu et al. 2018).

Relationship between commuters' departure time choices and the evolution of en-route traffic and a data linkage method was studied via a dynamic modeling framework in (Dong and Cirillo 2018). A regional household travel survey data was linked to travel information obtained from the Google Map application program interface (API), creating a synthetic longitudinal data set. Two decision rules, named perfect knowledge and stochastic changes, were applied to model the evolution of traffic. The results indicated that travel time, trip distance of commute trips, and commuters' sociodemographic influenced departure time choices.

A comparison of the reviewed departure time choice behavior research is summarized in Table 2 below, in chronological order. The comparison is based on the model developed, source of data (field data versus survey), size of data used (high, medium or low), provision of information or incentives, contributing factors it focused on, and the type of trips it investigated.

[Insert Table 2 Comparison of departure time choice behavior research here]

4. ROUTE CHOICE BEHAVIOR

The design principals of most popular Advanced Travel Information Systems (ATISs) such as Google Maps, Waze or other navigation apps are to assist travelers in making better route choices by providing dynamic traffic information. As such, lots of research attention has been paid on studying the impact of traffic information and other factors on traveler's route choice behavior, and various methodologies frameworks and experiments have been designed in previous research. For example, five models were built in (Abdel-Aty and FathyAbdalla 2006) to address drivers' diversion, compliance, and route choice under the effect of advanced traveler information systems. Dynamic data was collected by a travel simulator providing five different levels of traffic information/advice and collecting pre-trip and enroute route choices. The binomial and multinomial generalized extreme equations were used and the modeling results showed that expressway users were shown as the most travel-time savers who would divert if they were guided to a less-travel-time alternative.

The effects of pre-trip information on route-choice decisions when travel conditions were congested and stochastic was studied in (Lindsey et al. 2014). A model based on the classical two route network was adopted, in which free-flow travel times and/or capacities on each route varied unpredictably due to such shocks as bad weather, accidents, and special events. The results showed that the benefits of information depend on differences between routes in free-flow costs, the shape of the travel cost functions, the severity of congestion and capacity shocks, and the degree of correlation between routes in travel conditions. The developed model was tested in (Rapoport et al. 2014) in which an experiment was conducted. They reformulated the model for finite populations, and then tested and found support for its predictions in the experiment where under different capacity scenarios, and with and without pre-trip information, subjects were asked to choose routes with payoff contingent on their performance.

The impact of enroute real-time information about the occurrence of an incident (prevailing) and ex post information on forgone payoffs on traveler's route choice behavior was studied in (Lu et al. 2011). Data were collected from an interactive experiment in which participants made multiple rounds of route choices on a hypothetical network, and travel times were determined by performance functions of route flows from the previous round. Enroute real-time information increased the network's travel-time savings and reliability under the experimental setting, yet FP information had the opposite effect. Real-time information at downstream nodes encouraged participants' strategic behavior at the origin. FP information appeared to increase risk-seeking behavior; it encouraged route switching without real-time information and suppressed it with real-time information. The sensitivity of travelers' response to information, especially in the case of accepting route suggestions depended on their perceptions of the provided information's accuracy.

Drivers' behavioral responses to social navigation route guidance under different information and incentive strategies was empirically evaluated in (Djavadian et al. 2014). A traffic navigation application was developed and used in a pilot multiuser laboratory experiment, in which participants were asked to make route choices in a virtual travel environment under various information and incentive strategies. The results showed that drivers were more willing to comply with the social advice when they were well informed or rewarded, and female/novice drivers were more willing to comply with the social advice than were male drivers and experienced drivers. (Zhang et al. 2018) investigated the effects of social interaction information from friends on commuters' daily route choice decisions. The authors developed a day-to-day route-choice learning model with friends' travel information based on the Cumulative Prospect Theory (CPT), and then designed and conducted a laboratory behavioral experiment to collect the statistical data associated with subjects' actual route choice decisions. The results showed that a larger rate of social interactions in an online travel community does not necessarily lead to a better route-choice outcome for individuals or the whole system.

Several surveys and experiments were conducted to obtain the revealed preferences (RP) and stated preferences (SP) on route choice. (Shiftan et al. 2011) conducted a field study and an in-laboratory

experiment to obtain revealed preferences and stated preferences on route choice, and used a combined RP-SP route-choice model. The results suggested that when experience seeking increases, individuals tended to prefer a route characterized by lower average but greater variance of travel time. Other results confirmed that information had an impact on route-choice behavior although compliance with received information decreases with recent experience. (Kusakabe et al. 2012) conducted a stated-preference survey to describe drivers' route choice behaviors when traffic incident information was provided on a Variable Message Sign (VMS) in urban expressway networks. The results showed that travelers can assume travel time of their alternative routes according to the congestion information of the road section provided by VMS. (Razo and Gao 2013) took a Stated Preference survey conducted with interactive maps in order to analyze strategic route choice, where a traveler was able to plan ahead for traffic information that s/he will receive in the future. The analysis suggested that a traveler's risk attitude was probability-dependent, which was explained by a rank-dependent expected utility (RDEU) model. A latent-class mixed Logit model for panel data was specified with a RDEU component and two latent classes, strategic and non-strategic route choice. The estimated results suggested that a route choice model under real-time information should consider both types of behavior.

With the continuous improvements of the computation, communication, sensor technologies and the era of big data coming, the connected vehicle environment was born at the century and experienced a rapid development period since 2010. Drivers can move with the real-time traffic information under the connected vehicle environment. Meanwhile, individual driving behavior (e.g., route choice) can influence the traffic state of the connected vehicle. (Dai et al. 2018) investigated the relationship between the performance of traffic system and the driver route choice behavior in the connected vehicle environment. A dynamic process of commuter day-to-day route choice behavior under the connected vehicle environment was proposed, the driving experience and the real-time information that affect the driver route choice were taken into considered simultaneously. The authors conducted a series of simulation experiments under the different percentages of drivers who paid enough attention to fuel consumption, and then simulation results were compared. The simulation results demonstrated that the performance of

traffic state was better when enough drivers put emphasis on fuel consumption, and the travel time was more sensitive to the change of traffic state than fuel consumption.

A route-choice experiment was conducted in (Ben-Elia et al. 2013) to investigate the impact of information accuracy on route choice behavior. 36 participants, involving 20 repetitions under three different levels of information accuracy participated in the experiment. Aggregate analysis using non-parametric statistics and disaggregate analysis using a mixed logit choice model were applied. The results suggested that decreasing accuracy shifts choices not only from the riskier to the reliable route but also to the useless alternative. Prescriptive information had the largest behavioral impact followed by descriptive and experiential feedback information. (Liao and Chen 2015) applied the concept of a Bayesian game to the route choice problem when travelers were maximizing their degree of satisfaction by choosing routes, and a Bayesian model was used to analyze the impact of this heterogeneity on travelers' route choices. The results suggested that the use of ATIS exposes certain heterogeneities in traveler types and beliefs.

A learning-based model of route-choice behavior when information was provided in real time was presented in (Ben-Elia and Shiftan 2010). Participants made majority of binary route-choice trials relying on real-time information and learning from their personal experience reinforced through feedback in a laboratory-controlled experience. A discrete choice model with a Mixed Logit specification was estimated based on the experiment's data accounting for panel effects. The results showed that information and experience had a combined effect on drivers' route-choice behavior. Informed participants tended to base their decisions on memorization relating to previous outcomes with faster learning rates and had greater sensitivity to travel time variability.

Changes in traveler's route choice behavior in response to pre-planned events such as constructions or special events that lead to road capacity reductions and create bottlenecks in the traffic network was studied in (Hu et al. 2018). High-resolution vehicle trajectory data was collected via a smartphone app "Metropia", which was capable of keeping track of individual driver's behavior before and after road capacity reduction, to investigate travelers' behavioral responses to pre-planned events and the contribution factors. A functional data analysis (FDA) approach-based clustering method was proposed to

cluster trajectory data and identify detour patterns, and two logistic and a least absolute shrinkage and selection operator (LASSO) regression models were used to explain drivers' detour behavior choice for each pattern with spatial and temporal features of interest. A case study based on a lane closure event on MoPac expressway in Austin, TX was used as an example. The case study demonstrated that: 1) the freeway capacity reduction triggered heterologous behavior responses, 2) driver detour behavior exhibited three major patterns, and 3) each detour pattern highly depended on spatial features such as trip length, distance to freeway entrance and distance to other alternative freeways, in addition to the temporal features when the trip happens.

Apart from private vehicles, taxis also play an important role in providing on-demand mobility and their route choice behavior has also been extensively studied. Traditionally, vacant taxis cruise on roads searching for customers. In recent years, thanks to the proliferation of GPS-enabled smartphones, e-hailing applications (e.g., Uber, Lyft, and Didi Chuxing) are widely adopted by ride-sourcing drivers and in some cases, traditional taxi drivers to receive requests from nearby customers. The driver of a hired taxi usually aligns his/her routing objective with the passenger's, given the paramount importance of customer service (taking a detour to get higher fare is unprofessional and rare), and such a problem has been well studied. The interesting question is how to route vacant taxis. Taxis cruising on roads not only result in wasted gas and time for taxi drivers but also generate additional traffic in a city. Therefore, how to improve the utilization of taxis is of importance to both taxi drivers and the society. In an earlier study, (Hu et al. 2012) proposed a dynamic programming model of routing vacant taxis to depict the decisions at intersections according to the passenger arrival rate. Taxi drivers were assumed to minimize the expected search time for customers when making routing decisions at intersections. However, the expected search time was only minimized for the next customer, which might be inefficient in the long run. For this reason, experienced taxi drivers would not simply make their customer search decisions depending on the current searching time/profit, but would also consider the subsequent possible states that could be encountered. The majority of vacant taxi routing studies in the literature, in fact, only considers an optimization problem until meeting the next customer (e.g., (Qu et al. 2014); (Huang et al. 2015)), and in

some cases, the revenue from the next customer (e.g., (Yuan et al. 2013)). (Yu et al. 2018) formulated the vacant taxi routing problem as a Markov Decision Process (MDP) so that long-term objectives can be taken into account instead of the immediate one of meeting the next customer. The proposed model and solution algorithm in (Yu et al. 2019) were tested in a case study with parameters derived from GPS trajectories of over 12,000 taxis in Shanghai, China for a study period of 5:30 - 11:30 am on a typical weekday. The optimal routing policy was compared with three heuristics based on simulated trajectories. Results showed that the optimal routing policy improved average unit profit by 23.0% and 8.4% over the random walk and local hotspot heuristic respectively; and improved occupancy rate by 23.8% and 8.3% respectively.

A comparison of the reviewed route choice behavior research is summarized in Table 3 below, in chronological order. The comparison is based on the model developed, source of data (field data versus survey), size of data used (high, medium or low), provision of information or incentives, contributing factors it focused on, and the type of trips it investigated.

[Insert Table 3 Comparison of route choice behavior research here]

5. MODE CHOICE AND DESTINATION CHOICE BEHAVIOR

Mode choice and destination choice were found to change less as compared to departure time and route choice, according to a user satisfaction survey conducted in the Greater Cincinnati and northern Kentucky area (Aultman-Hall et al. 2000). Users were asked what portion of the time the traffic information affected their travel behavior. They reported that after the callers received information from a traffic advisory telephone service, 71 percent of the time their route taken cane be adjusted, and 34 percent of the time they can leave at a different time. On the other hand, they were willing to change their destination only for 25 percent of the time, and adjusted their mode merely for 5.7% of the time. The study also showed a total of 65 percent of users indicated they would be willing to pay for the service. On average, the maximum amount people would pay per call was $0.25.

(Abdel-Aty and FathyAbdalla 2006) collected and analyzed data that can be used to model mode choice paradigms using same subjects and same experiments. The effect of advanced traveler information systems (ATIS) on the mode choice was considered. A travel simulator was used as a dynamic data collection tool. The simulator used a realistic network, two modes of travel, real historical volumes, and different weather conditions. It provided five different levels of traffic information/advice, one at a time, and collected dynamic mode choices and pretrip (long-term) and enroute (short-term) route choices. The results showed that education level did not affect the mode compliance and providing pre-trip or enroute traffic information assisted travelers and drivers in their mode decisions. As the level of information/advice increased, the benefits increased. One interesting finding was that developing in-vehicle information devices may lead to a less transit usage in some cases, which indicated a potential drawback of this technology.

Chicago Metropolitan Agency for Planning (CMAP)'s Travel Tracker Survey, which is a household travel diary survey including both GPS and non-GPS components, was used to estimate a mode choice model for work trips to explore the influence of unreliability on travel behavior (Sweet and Chen 2011). The results suggested that unreliable auto travel conditions induce mode switching to transit and that the influence was strongest when service by train was already faster than by car. The results suggested that auto travel unreliability may have the strongest influence in metropolitan regions with highly-competitive transit systems, which triggered the mode choice change most effectively. Nevertheless, the influence of travel unreliability was limited and was not the underlying driver of travel decision-making.

Carsharing programs have been deployed in North America in 1994 and the organizations provided members access to a fleet of shared vehicles on an hourly basis, reducing the need for private vehicle ownership and encouraging members to make greater use of alternative modes of travel (Katzev 2003, Shaheen et al. 2009). (Katsev et al. 2001) analyzed the effect of CarSharing Portland (CSP) on travel behavior. The results showed that carsharing led to significant changes in the use of alternative transportation. After joining CSP, individuals took the bus more often, rode their bicycle more and did more walking than they had before. (Martin and Shaheen 2011) evaluated the travel patterns' shift

towards and away from public transit and non-motorized modes based on a 6281 respondent survey completed in late-2008 by member of major North American carsharing organizations. The results showed an overall decline in public transit use that was statistically significant, as 589 carsharing members reduced rail use and 828 reduced bus use, while 494 increased rail use and 732 increased bus use. At the same time, members exhibited a statistically significant increase in travel by walking, bicycling, and carpooling. Data collected on the commute distance of respondents found that carsharing members tend to have shorter commutes than most people living in the same zip code. (Costain et al. 2012) presented a comprehensive investigation of the behavior of carsharing members through the analysis of administrative datasets of a dominant carsharing program in Toronto. The dataset was collected from AutoShare, a Toronto based Carsharing Company, and included monthly transaction and membership directory data from 2008 to 2010. In 2010, a total of 84,773 trips were made by 6085 active members. Results showed that carsharing was most often used for off-peak period travel or on weekends, when transit service was poor and traffic congestion was low. The free-floating carsharing systems (FFCS) car2go (Daimler, 2009) and DriveNow (BMW, 2011) were offered by companies that possessed the organizational capabilities, technical knowledge and financial resources to roll-out carsharing systems large-scale (Firnkorn 2012). The new free-floating carsharing systems operated with more cars per city than most traditional carsharing organizations offer in total in all cities. (Kopp et al. 2015) analyzed the travel behavior based on substantial data that was collected with an innovative survey design based on a GPS tracking smartphone application. They contrasted the core group of members of FFCS model DriveNow (male, 25-45 years old) with people who did not use car sharing. The results showed that FFCS reported more trips, which covered shorter distances, and differed substantially in their mode choice behavior. It also demonstrated that FFCS were more intermodal and multimodal in their behavior and they choose bikes significantly more and motorized private transport modes less.

Models were built to understand the determinants of shopping destination choice in (Scott and He 2012). A total of 616 shopping trips were selected from a travel survey conducted in five counties of the Louisville KY-IN MSA in 2000, and shopping opportunities were aggregated to traffic analysis zones

(TAZs) for analysis. Multinomial logit models were developed to estimate the determinants of shopping destination choice and potential path area to determine an individual's destination choice set given their spatio-temporal constraints. In this way, the destination choice set was a more realistic representation of the shopping alternatives available to the individual. The results suggested that constrained models were better predictors of destinations than unconstrained models. The advances in GPS and GIS technologies provide new opportunities for collecting travel data.

A research investigated how land use, road network structure, and axis of travel influenced home-based, non-work destination choice based on in-vehicle GPS travel data in the Minneapolis-St. Paul Metropolitan Area in 2008 (Huang and Levinson 2015). In-vehicle GPS travel data was used to model non-work destination choice, and a new approach combining survival analysis and random sampling to form choice sets was proposed. The results suggested that a non-work destination closer to the axis of home and work was more attractive to auto users, a non-work destination closer to downtown was less attractive, and a non-work destination reached by a route with greater changes of speed per unit time or more turns per unit time was less attractive.

Given the real-time trip origin and destination information, (Huang and Levinson 2017) adopted mixed-effects logit models to model home-based trip chains with two destinations and collected in-vehicle GPS travel data which recorded 141 participants' vehicle trips including origins, parking destinations, and travel routes during a study period of five months. Three alternatives of building choice sets where various relationships of the two destinations (such as major–minor destinations, selecting one first, and selecting two concurrently) were considered, proposed and compared. The hypothesis had been tested that not only the spatial relationship but also the land use relationship of the destinations in a trip chain affect the decision making process. The results suggested that trip chaining behavior was shaped by the features of retail clusters, spatial patterns of clusters, transportation networks, and the axis of travel. It also reveals that not only the spatial relationship but also the land use relationship of the destinations in a trip chain affected the decision-making process.

6. DRIVING BEHAVIOR

Traditional driving risk analysis is usually performed based on crash history data or survey results, and uses this information to identify hazardous locations or driving risk factors contributing to the crash events. Research in this type reveals driving risk factors based on the correlation analysis at the aggregate level, and typically does not go down to the individual level due to the absence of personalized driving data. An emerging stream of research has been applying the latest Information and Communication Technology in data collection and driving behavior evaluation and adjustment in the last decade. Such data usually include not only vehicle spatial-temporal dimension location, but more importantly also speed, acceleration, and deceleration, which are frequently used to evaluate a driver's driving behavior.

6.1 The 100-Car Naturalistic Driving Study

The 100-Car Naturalistic Driving Study was the first instrumented vehicle study undertaken with the primary purpose of collecting large-scale naturalistic driving data. 100 participants (40 women and 60 men) between the ages of 18 and 64 years old from the Washington, D.C., area were recruited and In-Vehicle Data Recorders (IVDR) was instrumented to continuously measure the vehicle's position, speed and acceleration using GPS and accelerometers as well as video cameras. Drivers were given no special instructions, no experimenter was present, and the data collection instrumentation was unobtrusive. Four levels of driving events - crash, near-crash, incidents, and baseline - were classified from the video by trained video reductionists. The resulting database contained many extreme cases of driving behavior and performance, including severe drowsiness, impairment, judgment error, risk taking, willingness to engage in secondary tasks, aggressive driving, and traffic violations. The data set included approximately 2,000,000 vehicle miles, almost 43,000 hours of data, 241 primary and secondary drivers, 12 to 13 months of data collection for each vehicle, and data from a highly capable instrumentation system including 5 channels of video and many vehicle state and kinematic sensors (Dingus et al. 2006). Based on such a dataset, a list of studies was performed. Significant risk factors were identified in (Guo and Fang 2013) via a negative binomial regression model. High- and moderate-risk drivers were predicted

with two logistic models. The cluster analysis identified approximately 6% of drivers as high-risk drivers, with average crash and near-crash (CNC) rate of 3.95 per 1000 miles traveled, 12% of drivers as moderate-risk drivers (average CNC rate = 1.75), and 84% of drivers as low-risk drivers (average CNC rate = 0.39). The results indicated that crash and near-crash risk for individual drivers was associated with critical incident rate, demographic, and personality characteristics. Differences in braking behavior between age and gender groups in car-following scenarios were characterized in (Montgomery et al. 2014), in which the mixed model analysis of variance was used to examine the differences. The results showed that males age 18–20 had the lowest average minimum time to collision (TTC) at braking of 2.5 ± 0.8 s, and females age 31–50 had the highest average minimum TTC at braking of 6.4 ± 0.9 s. Age was found to be a statistically significant factor for TTC at braking between participants under 30 and participants over 30, with the latter braking 1.7 s on average before the former.

Klauer et al. also used such dataset to study the relationship between the performance of secondary tasks, including cell-phone use, and the risk of crashes and near-crashes (Klauer et al. 2014). A mixed-effects linear-regression model was used to assess trends for performance of a secondary task over time by both novice and experienced drivers. Results suggested that inattention contributed to 78% of all crashes collected over the 12-month data collection period, and specifically, the risk of a crash or near-crash among novice drivers increased significantly if they were dialing a cell phone, reaching for a cell phone, sending or receiving text messages, reaching for an object other than a cell phone, looking at a roadside object, or eating. Driver distraction was also studied in (Liang et al. 2012) which predicted crash and near-crash risk on the basis of driver glance behavior and demonstrated that distraction estimated from driver eye-glance patterns was positively associated with crash risk. A variety of publications also took advantage of such dataset and focused on other aspects of driving behavior, such as seat belt use (Reagan et al. 2013), lane-change behavior (Fitch et al. 2009), run-off-road crash (McLaughlin et al. 2009), real-end crashes and near-crashes (Jonasson and Rootzén 2014), and individual driver risk assessment (Guo and Fang 2013).

6.2 Pay-As-You-Drive-And-You-Save Insurance Actuarial Study

Traditional insurance models are based on a combination of static factors such as the driver's socio-demographic profile and vehicle information in conjunction with driving history. With the advent of new mobile technologies, some insurance companies have begun exploring a new auto insurance model known as Usage-Based-Insurance [UBI] or Pay-As-You-Drive-And-You-Save [PAYDAYS], which aims to incorporate individualized, real-world dynamic driving patterns into actuarial pricing. In 2013, the Federal Highway Administration contracted a research project to prove their PAYDAYS theory and study the implications of such an approach to insurance companies' premium calculation model. The motivation behind this project was that while most existing UBI or PAYDAYS efforts can be considered a major leap forward, the majority of researchers and insurers relied solely on user GPS trajectories, which only measured the specific driver's performance--such as the number of miles driven, travelling speeds, and hardness of braking--without considering other critical risk factors in the surrounding environment that may also contribute to crash risk, i.e. the contextual-sensitive risk factors. A total of 131,537 trips taken by 503 panel members over an 18-month period was collected in this study, which corresponds to 1,090,136 miles traveled. Individualized driving behavior data was collected from the driver's smartphone GPS module, and combined with geographical network information and dynamic traffic conditions to identify driving risk factors and evaluate driving behaviors under various contexts (Hu et al. 2018).

The benefits of utilizing context-relevant information in the driving behavior assessment process, i.e. contextual driving behavior assessment approach, were evaluated in (Zhu et al. 2017). A Bayesian Network (BN) model that investigated the relationships between GPS driving observations, individual driving behavior, individual driving risks, and individual crash frequency was developed. In contrast to prior studies without context information (i.e. non-contextual approach), the data used in the BN approach was a combination of contextual features in the surrounding environment that may contribute to crash risk, such as road conditions surrounding the vehicle of interest and dynamic traffic flow information, as

well as the non-contextual data such as instantaneous driving speed and the acceleration/deceleration of a vehicle. The analysis results showed that the contextual model had significantly better performance than the non-contextual model, and that drivers who drove at a speed faster than others or much slower than the speed limit at the ramp, and with more rapid acceleration or deceleration on freeways were more likely to be involved in crash events. In addition, younger drivers, and female drivers with higher VMT were found to have higher crash risk.

Hu et al. further examined the relationship between the driving performance measurements and accident rate, which was verified by combining the crash history of the drivers. A total of five measurements were defined to capture the driving performance at the trajectory and link level, including relative speed, contextual speeding, smoothness, hard brake and hard start. Based on those performance measurements, trip level performance was then aggregated for each trip from the trajectory and link level measures. Numeric analysis on the trip level demonstrated that driving behavior was context-sensitive, and the principal component analysis performed at the driver level showed that pedal operation and driving speed were the two most important performance measurements to characterize an individual's driving pattern. The subsequent correlation analysis of crash history and driving performances verified both pedal operation and driving speed were significantly related to more at-fault accidents (Zhu et al. 2013, Hu et al. 2015, Hu et al. 2018).

Ma et al. demonstrated how second-by-second GPS data can be integrated into existing or new auto insurance pricing structures in (Ma et al. 2018). Two types of data, real-time GPS trajectory data collected from the PAYDAYS project, as well as survey data were used. Vehicle trajectories and accident data were incorporated to quantify the relationship between driving hazard and accidents with the goal of establishing the linkage between driving risks and accident costs. GPS data considered in this study included not only tradition UBI factors, but also the unique contextual-based risk measurements that compared the driving speed of the vehicle with other drivers on the same road segment. It was found that hard brakes, hard starts, peak time travel, speeding as well as driving at a speed significantly different from traffic flow were highly correlated with accident rate. The potential underwriting loss an insurer may

incur resulting from adverse selection if omitting pertinent risk factors was illustrated. The results were helpful for insurance companies that are interested in getting into the UBI area to set up their auto insurance premium rates.

7. CONCLUSIONS AND DSICUSSIONS

This manuscript reviewed the state-of-the-art research on the impact of ICT on triggering traveler behavior changes. The use of dynamic traffic information and incentives, two most commonly seen means, are frequently used to effectively influence people's travel behavior. The review started from a review of system development and current practices, followed by behavior changes in the form of departure time choice, route choice, destination choice and mode choice, and driving behaviors. Transportation technologies have been and will continue to advance rapidly, and as such, traveler's pre-trip decision making and enroute experience will continue to be assisted with the new technologies. The transportation data to be collected will also be greatly expanded not only in volume, but also in velocity and variety. It will facilitate researchers in developing new and better models and algorithms to investigate traveler's behavior, and study what it takes to influence the decision-making process. The advancement of ICT will also facilitate the real-time or near real-time behavioral interactions, to design behavior experiments by presenting information and incentives/cost to the users, and observe how their behavior may adjust with a very short implementation cycle. Such efficient experiment design of having the human in the loop is another promising research direction to go. Last but not the least, how to further behavior change and make it a habit has not been well studied. While the offering of incentives has been proven to be effective, the hope is the behavior change can last even if the incentives are withdrawn, and the travelers have formed the habit of making informed and better choices of trip making. Interdisciplinary research combining traffic safety, psychology, economics and researchers from the other areas will be beneficial to advance behavior research to the exciting future.

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Table 1 Comparison of systems development and behavior experiments

Research	Date	System/Product development	Scale	ICT	Carrot or Sticks	Behavior choices
U-Pass program, UW	1991	No	Small	No	Incentives	Departure time, parking choice
Variable bridge toll, Lee County	1994	No	Midsize	No	Pricing	Departure time
SmartRide	1996	Yes	Midsize	Yes	N/A	Mode choice
NECO Pass Program	1997	No	Large	No	Incentives	Mode choice
Transit pass program, Portland	1999	No	Large	No	Incentives	Mode choice
Zion National Park	2000	No	Small	No	Incentives	Mode choice
One Less Cae Program	2000	No	Small	No	Incentives	Mode Choice
Pre-paid bus ticket, Germany	2003	No	Large	No	Incentives	Mode choice
Free bus ticket experiment, Japan	2003	No	Small	No	Incentives	Mode choice
NuRide	2003	Yes	Large	Yes	Incentives	Mode choice
Spitsmijden	2006	No	Large	Yes	Incentives	Departure time, mode choice, teleworking

INSTANT	2008	No	Large	No	Incentives	Departure time
Rail program, Melbourne	2008	No	Large	No	Incentives	Departure time
CAPRI	2012	No	Midsize	No	Incentives	Departure time, mode choice
Metropia	2013	Yes	Large	Yes	Incentives	Departure time, Route choice, mode choice;
Subway program, Singapore	2013	No	Large	No	Incentives	Departure time
Bella Mossa program, Italy	2017	Yes	Large	Yes	Incentives	Departure time
incenTrip	2018	Yes	Large	Yes	Incentives	Departure time, Route choice, mode choice;

Table 2 Comparison of departure time choice behavior research

Research	Date	Data source	Size of data	Type of trips	Provision of Information, incentives	Major contributing factors	Model developed
(Khattak et al. 1995)	1995	Survey	Medium	Home-to-work trips	Traffic reports	Personal characteristics	Probit model
(Bhat and Steed 2002)	2002	Survey	Medium	24-hour shopping trips	N/A	Socio-demographics, employment-related attributes and trip-related characteristics.	Continuous-time hazard duration model
(Saleh and Farrell 2005)	2005	Survey	Medium	24-hour work trips	N/A	Non-work activities and socio-economic characteristics	Multinomial logit model
(Gadda et al. 2009)	2009	Field	High	24-hour trips	N/A	Demographic and trip-related variables	Lognormal and weibull model
(Habib et al. 2009)	2009	Survey	High	24-hour H-W-H trips	N/A	Socio-demographic variables and trip attributes	Joint multinomial logit mode choice and

							Continuous time hazard model
(Ben-Elia et al. 2010)	2010	Field data	High	Peak hour trips	Incentives	Incentives, social demographics	Logit, mixture, latent class model
(Lemp et al. 2010)	2010	Survey	Medium	24-hour trips	N/A	Transport-related variables	Continuous cross-nested logit model
(Lemp and Kockelman 2010)	2010	Survey	High	24-hour trips	Travel time	Individual-specific variables	Continuous logit model
(Yang et al. 2013)	2013	Survey	Medium	24-hour shopping trips	Travel time	Socio-economic variables and individuals' preference	Continuous cross-nested logit model
(Thorhauge et al. 2016)	2016	Survey	Medium	24-hour out-of-home trips	Pricing	Work constraints, non-work activities	Mixed logit
(Thorhauge et al. 2016)	2016	Survey	Medium	Peak hour work trips	Experienced congestion	Psychological factors	Hybrid choice model

(Yang et al. 2016)	2016	Survey	Medium	24 hour work trips	N/A	Individual characteristics	Multinomial logit model
(Thorhauge et al. 2017)	2017	Survey	Medium	Peak hour trips	Toll	Attitude, social norms, and perceived behavioral control	Hybrid choice model
(Zhu et al. 2017)	2017	Survey	High	24-hour home-based trips	Travel cost	Social demographical variables	Mixed Bayesian network
(Shabanpour et al. 2017)	2017	Survey	High		N/A	Individuals' and households' demographic	Multinomial logit and log-linear regression model

Table 3 Comparison of route choice behavior research

Research	Date	Data source	Size of data	Provision of Information, incentives	Major contributing factors	Type of trips	Model developed
(Abdel-Aty and FathyAbdalla 2006)	2006	Field	Medium	Traffic information/advice	Age, gender, income, driving experience, and frequency usage of the expressways	Morning–week–day trips	Binomial and multinomial generalized model
(Ben-Elia and Shiftan 2010)	2010	Survey	Medium	Information	Travel time variability, risk-taking	24-hour trips	discrete choice model
(Lu et al. 2011)	2011	Survey	Small	Ex post information	Average travel times, route shares, and route switches	Work to home trips	Block bootstrap method
(Shiftan et al. 2011)	2011	Survey	Medium	Pre-trip travel time information	Drivers' personal characteristics	Morning peak hour trips	Combined RP – SP route-choice model
(Kusakabe et al. 2012)	2012	Survey	Medium	Traffic incident information	Off-ramp choice behavior	24-hour trips	Multinomial logit model

(Razo and Gao 2013)	2013	Survey	Medium	Traffic information	Travelers' risk attitudes	24-hour trips	Latent-class mixed Logit model
(Ben-Elia et al. 2013)	2013	Survey	Small	Information	information accuracy	24-hour trips	mix-logit model
(Lindsey et al. 2014, Rapoport et al. 2014)	2014	Survey	High	Pre-trip information	Travel condition	Peak-hour trips	Experience-Weighted Attractions learning model
(Djavadian et al. 2014)	2014	Survey	Medium	Incentives	social navigation route guidance	From Delft or Gouda to Dordrecht	Dynamic descriptive behavioral model
(Liao and Chen 2015)	2015	Survey	Small	Information	Travel time	24-hour trips	Bayesian game
(Zhang et al. 2018)	2018	Survey	Medium	Friends' travel information	Social interactions	Daily commute trips	Cumulative Prospect Theory-based day-to-day route-choice model

(Dai et al. 2018)	2018	Survey	Small	Information	fuel consumption, traffic condition	24-hour trips	linear utility maximization weighted model

Figure Legends:

1. Figure 1 Metropia smartphone app workflow
2. Figure 2 (a) incenTrip architecture
3. Figure 2 (b) incenTrip workflow